# The Run Budget: Ascent as a 2-adic Clock

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**Abstract**

For the accelerated Syracuse map, ascent occurs exactly at valuation $b = 1$, and we show the lengths of ascending runs are not statistics but *deterministic readings of a 2-adic clock*: a maximal $b$=1 run starting at $x$ has length exactly $v_2(x+1) - 1$, the ghost depth of $x$, which decreases by precisely one per step within the run. Consequently every divergent orbit must acquire its infinite height through visits to deep ghost points $x \equiv -1 \bmod 2^g$, with each visit capped by $g \le \log_2(x+1)$, and the quenched exclusion problem for band crossings reduces to a single renewal coordinate: the empirical law of the run-start depth against its natural density $2^{-g}$.

**Lemma 1** (Ascent = depth clock)**.** *Let $x$ be odd, $g = v_2(x+1)$. Then:* (i) *$b(x) = 1$ iff $g \ge 2$;* (ii) *if $b(x) = 1$ then $v_2(S(x)+1) = g - 1$, since $S(x) + 1 = \frac{3x+1}{2} + 1 = \frac{3(x+1)}{2}$ and 3 is odd;* (iii) *hence the maximal $b$=1 run starting at $x$ has length exactly $g - 1$, gaining $(g-1)\log_2 \frac{3}{2} + O(g/x)$ bits of height, and terminating at a point $\equiv 1 \bmod 4$;* (iv) *runs are capped: $g \le \log_2(x+1)$, so a single run at height $Y$ gains at most $\log_2(Y+1) \cdot \log_2 \frac{3}{2}$ bits. (Machine-checked: 20,000/20,000 runs, 20,071/20,071 single-step depth decrements, exact.)*

*Proof.* (i) $b(x) = 1 \iff 3x+1 \equiv 2 \bmod 4 \iff x \equiv 3 \bmod 4 \iff g \ge 2$. (ii) as displayed. (iii) iterate (ii) until depth 1, i.e. $x \equiv 1 \bmod 4$, where $b \ge 2$. (iv) $2^g \mid x+1$ and $x+1 > 0$. □

**Theorem 1** (Run budget for crossings)**.** *Let an orbit segment cross $[Y, 2^{2D}Y]$ upward with no descent below $Y$. Writing $g_1, g_2, \ldots$ for the ghost depths at its run starts and $d_1, d_2, \ldots$ for the (negative) drifts of the intervening $b \ge 2$ steps, the crossing satisfies*

$$\log_2 \tfrac{3}{2} \sum_i (g_i - 1) \;\ge\; 2D + \sum_j |d_j| - O\Big(\tfrac{M}{Y}\Big), \qquad g_i \le \log_2(2^{2D}Y + 1).$$

*In particular a crossing in $m$ steps at speed within $(1+\delta)$ of the minimum $1/(\log_2 3 - 1)$ forces the average run-start depth $\overline{g} \ge c(\delta)^{-1}$, i.e. fast crossings oversample deep ghost points relative to the natural density $2^{-g}$ by an exponential factor.*

*Proof.* Sum the height identity over the segment, splitting steps into runs (Lemma 1(iii)) and run-enders; the cap is Lemma 1(iv). □

**Corollary 1** (Target 2, reduced to one coordinate)**.** *A divergent orbit must satisfy $\sum_i (g_i - 1) = \infty$ with every $g_i \le \log_2(x_{\text{start},i} + 1)$: its run-start depth process must persistently exceed the annealed law (*$\Pr[g = j] = 2^{-j+1}$ *on odd integers, mean run length 1). The quenched exclusion problem for band crossings is thereby reduced to a single renewal coordinate:* no positive orbit may oversample its own ghost depths forever*. This is the 2-adic, single-coordinate analogue of the shadow-bias exclusion, and the natural meeting point with the phase-diagram confinement: windows allow no bias, so the bias — and now, explicitly, the depth oversampling — must live in crossings.*

**Remark.** *The depth process is observable, cheap, and sharply structured: depth at a run start is read off the binary expansion of $x+1$, the run then counts it down deterministically, and the next depth is determined by the 2-adic digits of the run-end image. The empirical comparison of orbit depth-laws with $2^{-g}$ (see `search/depth_stats.py`) is the Target-2 experiment to run at scale.*